

Exploring Virtual Reality for Religious Education in Real-World Settings

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Figure 1: The two VR setups during their deployment in a real-world setting. Left: Blessed Spaces. Right: VR Pastor.

ABSTRACT

Research and design of virtual reality (VR) applications for educational contexts often focus on science-related subjects and evaluate knowledge acquisition while overlooking other subjects like religious education or ‘what actually happens’ when VR is used in real-world settings. Our article combines both and presents two VR applications, *Blessed Spaces* and *VR Pastor*, designed for individual experiences in (Protestant) religious education. We deployed the applications in a real-world setting, an out-of-school learning centre. Surprisingly, the applications mediated social engagement between students. Our findings challenge traditional notions of social experiences in VR-supported education and call for more research in real-world settings.

Index Terms: Human-centered computing—HCI design and evaluation methods; Virtual Reality;

1 INTRODUCTION

As VR head-mounted displays (HMDs) become increasingly affordable, VR enters the everyday lives of children and young people. This target group is also focal to the constantly growing branch of VR for educational purposes. Researchers expect many positive effects on learning, such as making the abstract experienceable or distant places more accessible [8]. Existing research and applications of VR for educational purposes focus on subjects such as science, technology, and maths [3] or space, nature, and medicine [9]. In turn, subjects such as religious education have received less attention. Religious education means initiating processes of exploring religious cultures and their meanings. Nowadays, this includes both analogue and digital cultures (*digital religion*). Thus, religious education has a particular demand for integrating interactive technologies such as VR. However, this integration has not yet been achieved satisfactorily in everyday school life as interactive technologies tend to be perceived as disruptive in religious education [5]. New ways of practically integrating interactive technologies into religious education are being sought.

However, most educational VR research focuses on the ‘state of the art’, assessing knowledge acquisition and speculating about how technology should or could be [1]. This focus neglects what

happens when technology is actually deployed in real-world settings with their contextual factors and social dynamics [7, 10]. If research findings are to be translated into practice, research on the ‘state of the actual’ is needed [10]. Here we report from an interdisciplinary project between Protestant religious education and HCI to develop VR applications for religious education and explore how students interact with those applications when deployed in real-world settings.

2 VR APPLICATIONS FOR RELIGIOUS EDUCATION

To contextualise our work and make it accessible for religious education, we concentrated on interreligious blessing spaces as a topic specifically relevant to religious education. We familiarised ourselves with the topic of blessing through surveys and literature research (reported elsewhere [4]). To design respective VR applications, we performed several ideation workshops with researchers from religious education and HCI. The workshops started with a short introduction to blessing and religious education, followed by several rounds of idea generation supported by cards for ideating novel interaction technologies [2]. This ideation process resulted in various ideas, including ideas that had no connection to VR (reported elsewhere [4]). We clustered the ideas and developed them further through iterative prototyping and testing with researchers from religious education, HCI, and design. Ultimately, we implemented two VR applications, named *Blessed Spaces* and *VR Pastor*.

Blessed Spaces consists of two phases, experience and reflection. In the experience phase, students visit various places of blessing in VR, e.g., a beach by the sea where the waves gently lap against the shore. Comparing the experiences in VR with previous experiences should initiate reflections on own perspectives: What is a blessed space for me? What are the special aspects of blessed spaces? Back in reality, students entered the reflection phase and were asked to record their thoughts on a large sheet of paper and thus contribute to a (class) collage on blessed spaces. The application was intended to enable individual experiences allowing students to visit different places of blessing. To this end, we implemented a Unity custom application deployed to a Pico Neo 3 Pro. The six blessing places changed automatically after a while (30-50 seconds), with slow transitions and black fades to avoid nausea. In addition, students should remain seated during the experience. We selected the places based on results from our survey on blessing experiences [4].

VR Pastor is based on the Aaronite blessing, the pastoral blessing at the end of Christian worship services in which a pastor speaks blessings with raised arms. As many students today are unfamiliar with this blessing ritual, the application was made to support its

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understanding. To this end, students slip into the role of a pastor. The role change is initiated in reality, where students can put on a pastoral robe if they wish (see Figure 1). In VR, the students find themselves behind an altar, looking out over a church congregation. The voice of a randomly selected male or female pastor welcomes them and explains the context, elements, and functions of the Aaronite Blessing. The students are then free to decide whether to give an Aaronite blessing themselves. If they do, the blessing becomes visible through bright lights streaming from them towards the congregation. Taking on the pastoral role not only allows a change of perspective but also a critical scrutiny of the pastoral role. The experience was designed for individual use and implemented in Unity. To make interaction as natural as possible, we implemented gesture recognition using a Microsoft Kinect (e.g., hands raised for blessing). The accompanying texts were spoken by a male and female pastor, who each explained the Aaronite blessing from their perspective.

3 DEPLOYMENT OF VR APPLICATIONS IN THE REAL-WORLD

We decided to explore the applications' use in a real-world setting. It is clear from previous research that the direct use of VR within a school setting is accompanied by many hurdles (e.g., organisational, personnel, spatial) [10] and that interactive technology more generally might be perceived as disruptive in religious education classrooms [5]. Therefore, we propose an alternative approach: The use of new interactive technologies at an out-of-school learning centre that is technically supervised and can be visited during a field trip. We cooperated with the Hamburg Youth Church to make the VR applications publicly accessible as part of an interactive exhibition on blessing. This exhibition included other interactive exhibits such as a DJ desk for remixing interreligious blessings or a blessing booth for catching blessings blown through the air [4]. During one month, 200 people (school classes and confirmand groups) visited the exhibition. The settings were designed to minimize the risk of injury, allow individual experiences, and hide the technology irrelevant to the students (e.g., workstations, cables). Therefore, we used room dividers, colour coding (e.g., to indicate preferable positions), seating, and trained staff to supervise the exhibition. As space and resources allowed, we made Blessed Spaces available via three HMDs simultaneously so students, back in reality, could enter the reflection phase together (see Figure 1).

We carried out anonymised observations supported by protocol sheets, focusing on specifics in the interaction and physical, social, emotional, and intellectual engagement with the VR applications [6]. As part of the overall exhibition, introductory and reflection talks were held with discussions about expectations, preferences, and the topic of blessing itself, which we recorded through note-taking.

The initial analyses of the observations show some exciting results. For example, students expressed great enthusiasm for the VR exhibits. Positive comments focused on VR's stimulating and emotional effects, mentioning that it was particularly positive 'as I had never done this [VR] before and it was a great experience' or because it created a 'beautiful atmosphere' or was 'something novel'. Negative comments mostly concerned technical or organizational aspects. For example, the VR Pastor exhibit sometimes lost body tracking and long queues formed in front of the VR exhibits and students had to wait. This was also reflected in our experience: Compared to all other exhibits, the VR exhibits were the most maintenance-intensive. Although we extensively tested the VR applications in advance, unexpected problems arose during the exhibition (e.g., inconsistent lighting conditions influenced device- and body-tracking). Such incidents confirmed that using VR for educational purposes in an external, technically supervised location was valuable instead of deploying it in schools where teachers lacked technical support.

The most interesting finding was how social engagement occurred at both exhibits. Although both were designed to allow for individual VR experiences, they mediated social engagements between students.

For the VR Pastor exhibit, 40 % of all observations documented social settings, observing social engagements between the student in VR and bystanders, who asked about the experience or laughed together. In the Blessed Spaces exhibit, we observed social engagement in 100 % of our observations, and most of the time, the three students in VR interacted with each other, although each was in their own VR environment. For example, they discussed what they saw and how they experienced it. The spatial arrangement of the exhibit might have invited this unusual social engagement because the stools were close together, facing each other. Overall, these observations show that a social experience can arise not only when several people meet in VR but that other constellations are also possible.

4 CONCLUSION

Our work addresses gaps in the literature on VR for educational contexts in the humanities, such as religious education, and provides knowledge about the use of VR in real-world settings. We developed two VR applications for religious education, Blessed Spaces and VR Pastor, and tested them in a real-world setting, an out-of-school learning centre. Although we faced spontaneous, setting-related problems and the applications were designed to enable individual experiences in VR, we observed a high level of social engagement between the students at both VR exhibits, even while they were in separate virtual environments. This extended understanding of social experiences in VR-based educational scenarios opens up a new perspective on what 'social experience' in VR might mean and challenges to place social aspects and research in real-world settings more at the centre of future research.

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